

Bamboo Rhyme Craft: Immersive VR Bamboo Weaving through Coordinated Bimanual Gesture Interaction

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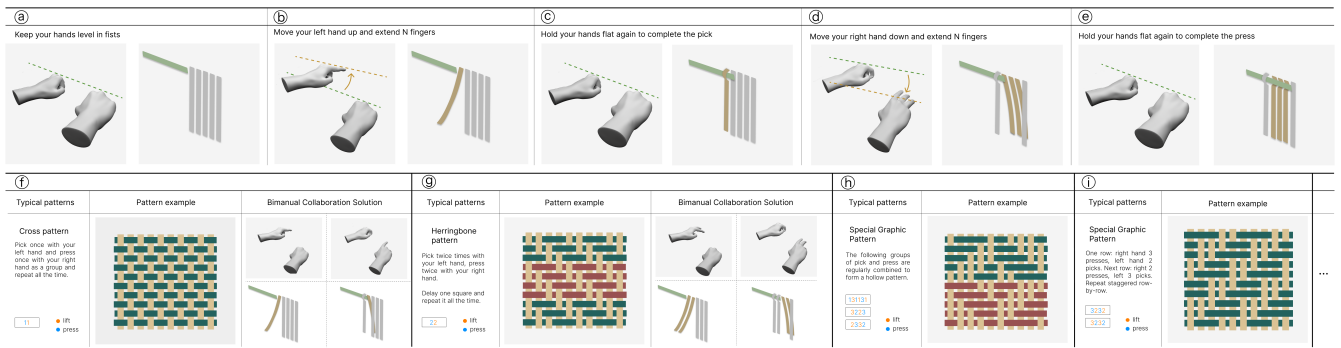


Figure 1: (a),(b),(c),(d),(e) Universal Bimanual Weaving Interaction: Exemplified by the "Left-hand Lift-One, Right-hand Press-Three" Technique(f),(g),(h),(i) Pattern Demonstration: Cross Pattern, Herringbone Pattern, Concentric Squares Pattern, Trapezoidal Pattern

Abstract

This research, developed in collaboration with inheritors of intangible cultural heritage (ICH) bamboo weaving, digitally translates traditional Chinese bamboo weaving techniques into a virtual reality (VR) gesture recognition interaction system. In this system, the user's hands act as proxies for bamboo strips, enabling the weaving process through coordinated hand gestures. The system allows users to clearly understand the structural principles of bamboo weaving while offering a novel and engaging gestural interaction experience. It also opens new possibilities for integrating bamboo weaving into the domain of virtual reality art, contributing to the broader dissemination and preservation of this traditional craft.

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CCS Concepts

• **Human-centered computing** → *Virtual reality; Interaction design theory, concepts and paradigms.*

Keywords

Gesture Recognition, Intangible Cultural Heritage, Bamboo Weaving, Virtual Reality, Interaction design

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1 Introduction

Bamboo weaving is one of China's intangible cultural heritage (ICH) traditions, with a history that spans several millennia. Integrating digital technologies with traditional weaving practices opens new possibilities for immersive crafting experiences. [Zhang et al. 2023] developed a VR-based bamboo weaving experience system and evaluated the effectiveness of its instructional modules. [Gao

et al. 2023]constructed a parametric design platform for digital bamboo weaving patterns analyzing the structural logic of traditional motifs.[Wu, Jue et al. 2021]created a motion-sensing interactive bamboo weaving experience using the Unity platform.[Yen et al. 2024]mapped hand joint tracking and gesture recognition to traditional puppetry performance gestures. Existing digital bamboo weaving experiences continue to focus primarily on parametric pattern design, with limited attention paid to gesture-driven interactions rooted in ICH practices. This research leverages the high-precision hand-tracking capabilities of the Pico 4 Pro to explore a gesture-based, immersive digital experience of traditional bamboo weaving techniques.

2 Gesture Interaction Process

This research integrates gesture recognition technology with the experiential knowledge of bamboo weaving inheritors to construct a bimanual gesture-based interaction paradigm for bamboo weaving. Through empirical evaluation by an intangible cultural heritage (ICH) inheritor of bamboo weaving, the proposed interaction model effectively captures the motion trajectories of the two core techniques—"lift" and "press"—at the level of action semantics. The model demonstrates significant advantages in both operational intuitiveness and cultural fidelity. During interaction, the user begins with both fists held level to establish a reference plane (Fig 1.a). To execute the "lift" gesture, the left hand is raised vertically above the right hand to create a height differential while extending "n" fingers (where "n" ranges from 1 to 5, and is extensible). The number of extended fingers directly maps to the number of vertical bamboo strips, triggering their corresponding animated deformation (Fig 1.b). Simultaneously, the right hand remains in a fist posture to provide spatial reference. When the left hand returns to the baseline plane, the system automatically completes the weaving of "n" vertical bamboo strips along with a horizontal displacement of "n" units (Fig 1.c). In contrast, the "press" gesture is executed by the right hand lowering and extending "n" fingers (Fig 1.d), then returning to the baseline and retracting (Fig 1.e). This design utilizes a dual-parameter mechanism—finger extension and spatial displacement—to achieve a digital translation of the traditional bamboo weaving technique characterized by the alternating "lift one, press one" method. The article further breaks down the weaving processes of cross pattern (Fig 1.f), herringbone pattern (Fig 1.g) and the special patterns such as concentric squares patterns (Fig 1.h) and trapezoid pattern (Fig 1.i) (Numeric codes at lower-left of patterns denote left/right hand press/lift counts).

3 Interaction Flow Design

The interactive experience is divided into two main sections. The first section provides a simplified animated tutorial, featuring a 4×4 grid of dynamic bamboo strips that guides users through gesture-based weaving operations to assist in the formal weaving process (Fig 2.a). The second section offers a comprehensive bamboo weaving interaction experience, structured into three phases. In the first phase, users engage in the practice of the weaving technique using bimanual gestures to weave a 10×10 grid of dynamic bamboo strips. (Fig 2.b). The second phase focuses on coloring the woven bamboo strips, where users can apply different colors to individual

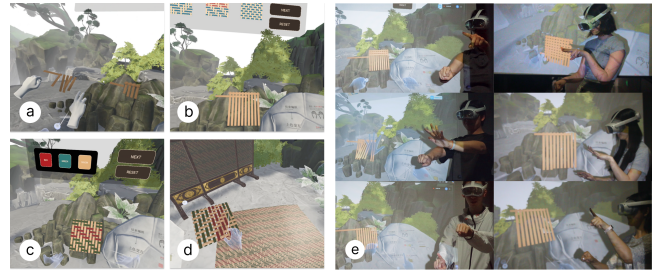


Figure 2: (a)Gesture Animation Tutorial(b)Pattern Weaving(c)Pattern Coloring(d)Pattern Application(e)Experience Record

strips by performing pinch gestures on color block UI elements and preview the results (Fig 2.c). In the third phase, three common bamboo woven product models—namely, a rug, a folding screen, and a lantern—are provided within the scene. Users can drag and drop their woven patterns onto these objects to preview the decorative effects (Fig 2.d). Every step above offers corresponding sound feedback based on user behavior.

4 Preliminary Feedback

The testing shows that the innovative interaction design not only reproduces the core processes of traditional bamboo weaving but also enhances user engagement through immersive experiences. ICH inheritors noted that the system captures the essence of traditional craftsmanship while appealing to younger audiences. They suggested adding a tangible weaving reward mechanism to strengthen the sense of achievement. These insights guide future iterations toward refining interaction details and developing the reward system to deepen the cultural transmission of ICH.

Acknowledgments

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